History, Training Coverage, and Action Chunking

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Abstract

Simchowitz et al. construct controllers that agree on demonstrations but fail outside the demonstrated states. We combine this example with history-based reconstruction and chunk-stability results. We evaluate the original two-state pair and coupled systems with 2, 4, 8, and 16 states. Noisy training actions supply corrective examples; with sufficient history, one-step control nearly matches a known-model observer. Noiseless training can still benefit from longer action execution. All test rollouts are noiseless.

1. The problem and theoretical results

The hard example of Simchowitz et al. (2025) pairs two systems, A_1, A_2, with stabilizing experts K_1, K_2. Both experts generate identical demonstrations when $x_1 = 0$, but need different corrections when $x_1 \neq 0$. Even perfect demonstration fit cannot distinguish their recovery rules. We fix A_1 and expert K_1: applying K_2 matches the demonstrations but destabilizes A_1 after a small departure. We ask whether past observations let a learner act accurately one step at a time instead of executing an action sequence before observing again.

Consider $x_{t+1} = Ax_t + Bu_t, y_t = Cx_t$, and a deterministic expert $u_t^* = Kx_t$, with $F = A+BK$ Schur stable. The learner receives $z_t = \mathrm{col}(y_{t-r+1:t}, u_{t-r+1:t-1})$, predicts H actions, and executes $\ell \leq H$ before replanning. Let h be the smallest integer with $\mathcal{O}_h = \mathrm{col}(C, CA, \ldots, CA^{h-1})$ full column rank. Norm $\|z\|_{\mathcal{Z}}$ sums Euclidean norms of history blocks. We separate representation, stability, and learning.

Proposition 1 (history can represent the expert). On noiseless, dynamically consistent histories, exact recovery of the current expert action is possible if and only if

$$\ker \mathcal{O}_r \subseteq \ker(KA^{r-1}).$$

In particular, $r \geq h$ permits a linear state reconstruction $x_t = R_r z_t$. The chunk policy $\pi_k^*(z) = KF^k R_r z$ reproduces the expert for every execution length $1 \leq \ell \leq H$.

Proof. With $s = t - r + 1$, stacked observations Y, and applied inputs U, write $Y = \mathcal{O}_r x_s + \mathcal{T}_r U$ and $x_t = A^{r-1}x_s + \mathcal{R}_r U$. Identical histories leave ambiguity $d \in \ker \mathcal{O}_r$, producing action ambiguity $KA^{r-1}d$. Kernel inclusion is therefore necessary and sufficient. For $r \geq h$, left inversion gives

$$R_r = [A^{r-1}\mathcal{O}_r^\dagger \quad \mathcal{R}_r - A^{r-1}\mathcal{O}_r^\dagger \mathcal{T}_r].$$

Applying $KF^k x_t$ then yields $x_{t+k} = F^k x_t$ by induction. □

Reconstruction uses known dynamics, actual inputs, and consistent noiseless histories. This existence result does not imply identification from demonstrations: expert histories can have low rank, and more coordinates need not add independent training information.

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1.1. History-based chunk stability

The next result allows nonlinear plants and general Lipschitz policies. Assume all admissible trajectory pairs satisfy the history-based exponential incremental input-to-output bound

$$\|\delta y_{t+k}\| \le C_0\rho^k\|\delta z_t\|_{\mathcal{Z}} + C_0\sum_{i=0}^{k-1}\rho^{k-1-i}\|\delta u_{t+i}\|, \tag{1}$$

with $C_0 \ge 1$ and $0 < \rho < 1$. Observable Schur-stable linear plants satisfy this bound for $r \ge h$, using reconstruction and exponential bounds on A^k. A state-based incremental stability bound alone also needs reconstruction to imply (1).

At chunk boundaries $t_j = j\ell$, apply

$$u_{t_j+k} = \pi_k(z_{t_j}) + v_{t_j+k}, \qquad \mathrm{Lip}_{\mathcal{Z}}(\pi_k) \le D\rho^k.$$

The offsets of π_k are unrestricted; the decay requirement concerns sensitivity to context. Both compared trajectories use the same policy and chunk schedule.

Proposition 2 (sufficient chunk length for incremental stability). Fix $\alpha \in (0,1)$ and define

$$\bar\rho = \rho^{1-\alpha}, \quad C_r = 2rC_0\rho^{-(r-1)}, \quad \bar C = C_0\left(1 + \frac{D}{e\alpha\rho\log(1/\rho)}\right).$$

If $\ell \ge \max(1, r-1)$ and

$$C_r(1 + D\ell/\rho)\rho^{\alpha\ell} \le 1, \tag{2}$$

then, writing $\widetilde C = \bar C C_r$,

$$\|\delta y_t\| \le \bar C\bar\rho^t\|\delta z_0\|_{\mathcal{Z}} + \widetilde C\sum_{i=0}^{t-1}\bar\rho^{t-1-i}\|\delta v_i\|. \tag{3}$$

Proof. Let $Z_j = \|\delta z_{t_j}\|_{\mathcal{Z}}$ and $V_i = \|\delta v_i\|$. Within a chunk, $\|\delta u_{t_j+k}\| \le D\rho^k Z_j + V_{t_j+k}$. Substitution into (1) gives, for $0 \le k \le \ell$,

$$\|\delta y_{t_j+k}\| \le C_0\rho^k(1 + Dk/\rho)Z_j + C_0\sum_{i<k}\rho^{k-1-i}V_{t_j+i}. \tag{4}$$

Since $\ell \ge r-1$, every block of $z_{t_{j+1}}$ lies in this chunk. Sum its r observation bounds and $r-1$ action bounds. Their coefficients on Z_j total at most

$$r\rho^{\ell-r+1}\{C_0(1 + D\ell/\rho) + D\} \le C_r\rho^\ell(1 + D\ell/\rho).$$

Each disturbance coefficient is at most $(rC_0+1)\rho^{\ell-r-i} \le C_r\rho^{\ell-1-i}$, including inputs earlier than the retained history. Consequently,

$$Z_{j+1} \le \bar\rho^\ell Z_j + C_r\sum_{i=0}^{\ell-1}\rho^{\ell-1-i}V_{t_j+i},$$

where (2) bounds the first coefficient. Iteration and $\rho \le \bar\rho$ imply

$$Z_j \le \bar\rho^{t_j}Z_0 + C_r\sum_{i<t_j}\bar\rho^{t_j-1-i}V_i.$$

Finally $\sup_{k\ge0} k\rho^{\alpha k} \le [e\alpha\log(1/\rho)]^{-1}$ gives $C_0\rho^k(1 + Dk/\rho) \le \bar C\bar\rho^k$. Insert the boundary estimate into (4), and absorb current-chunk disturbances using $C_0 \le \widetilde C$. This yields (3). □

1.2. From expert-context error to trajectory error

Corollary 3 (finite-horizon squared-loss bound). Assume Proposition 2 and identical initial histories. On a reference expert rollout define the chunk residual

$$e_{t_j+k} = u^*_{t_j+k} - \pi_k(z^*_{t_j}), \quad 0 \le k < \ell.$$

For T actions let $J_{\text{demo}} = T^{-1}\sum_{i<T}\|e_i\|^2$ and let $J_{\text{traj}} = T^{-1}\sum_{t=1}^{T}\|y_t - y^*_t\|^2$. Then

$$J_{\text{traj}} \le \left(\frac{\widetilde{C}}{1-\bar{\rho}}\right)^2 J_{\text{demo}}.$$

Proof. View the expert as the same chunked policy driven by $v_i = e_i$, and the learner as driven by $v_i = 0$. The initial term in (3) vanishes. The convolution kernel $\widetilde{C}\bar{\rho}^k$ has sum $\widetilde{C}/(1-\bar{\rho})$; the discrete ℓ_2 convolution inequality gives the result after squaring and dividing by T. □

This loss uses reference deployment histories, not empirical training data; substituting training loss requires a coverage/generalization argument. Averaging yields an expected bound. Under partial sensing, output error does not automatically control the full-state MSE below.

Proposition 2 does not prove shorter execution with more history: C_r grows with r, and it cannot certify $(r, \ell) = (16, 1)$. Its Lipschitz assumptions remain unverified for our learners. The simulations illustrate mechanisms rather than validate the bound.

2. Simulations: experiment, data, and results

The systems. We retain the original Simchowitz et al. (2025) pair at $n = 2$, $\mu = 1/8$, and separately construct a coupled family at $n = 2, 4, 8, 16$. These are not repeated 2D blocks. The first plant has $x_1^+ = 1.02x_1 + 0.04x_n + u_1$, $x_2^+ = -x_1 + u_2$, and $x_j^+ = x_{j-1} + u_j$ for $j \ge 3$, forming one feedback loop. The second replaces the first column of A_1 by $-0.995e_1$. With $F = \text{diag}(0, \ldots, 0, 0.75)$ and $K_i = F - A_i$, each expert stabilizes its own stable plant. Both give identical demonstrations on $\text{span}(e_n)$, but the wrong expert gives eigenvalue $+2.015$ or -2.015. Only x_n is observed; exact state and expert-action recovery need $h = n$ observations, with perfectly conditioned $\mathcal{O}_n$. The original and coupled $n = 2$ matrices differ and are never pooled.

Training and testing. Noiseless training follows expert trajectories on the demonstrated line. Noisy training adds known Gaussian changes of standard deviation 0.05 to applied actions, then labels visited states with exact expert continuations $K_i F^j x$. Noise changes the states visited, not the target labels; histories retain actual applied actions. This adapts noisy collection and expert correction (Laskey et al., 2017; Ross et al., 2011), motivated by Zhang et al. (2025).

Our learner is ridge regression ($\lambda = 10^{-6}$), not a procedure prescribed by the hard example. Five seeds each supply 128 trajectories of 100 steps. Target times match across history and prediction length. Single-action fits use $H = \ell = 1$; sequence fits use $H = 16$ and $\ell \in \{1, 2, 4, 8, 16\}$. Histories cover 1 through $n + 1$, plus common lengths 2, 4, 8, and 16. At fixed history the same sequence predictor is reused across executions; all first-action controls match the separate single-action fits.

After a common 16-step expert initialization, 40 held-out visible amplitudes each receive a hidden change of total magnitude 0.0005–0.0015, either in x_1 or distributed across hidden coordinates. At $n = 2$ both occupy the same axis. Test MSE averages squared full-state distance from a separate

expert rollout over 200 steps, both change types, and seeds. There are no continuing disturbances or expert queries. The original $n = 2$ comparison uses exactly the same protocol and initial states as coupled $n = 2$. Exact spectral stability is checked separately; selected 1000-step examples use matched initial states.

System	n	Training data	History r	Prediction horizon H	Execution horizon ℓ	Mean test MSE	Stable seeds
Original	2	Noiseless	2	1	1	6.09×10^{9}	0/5
Original	2	Noisy	2	1	1	7.07×10^{-9}	5/5
Coupled	2	Noiseless	2	1	1	5.04×10^{-6}	5/5
Coupled	2	Noisy	2	1	1	1.75×10^{-8}	5/5
Coupled	2	Noisy	2	16	16	1.36×10^{-7}	5/5
Coupled	4	Noisy	4	1	1	4.51×10^{-8}	5/5
Coupled	4	Noisy	4	16	16	2.05×10^{-7}	5/5
Coupled	8	Noisy	8	1	1	1.49×10^{-7}	5/5
Coupled	8	Noisy	8	16	16	3.90×10^{-7}	5/5
Coupled	16	Noisy	16	1	1	5.47×10^{-7}	5/5
Coupled	16	Noisy	16	16	16	5.54×10^{-7}	5/5

Selected settings; noiseless/noisy refer to training data. All testing is noiseless.

The results. With noisy training data, coupled-family single-action control at $r = h$ is stable in all 20 seed/dimension cases and nearly matches the known-model observer. At $r = h - 1$, all 20 fits are unstable. This supports Proposition 1's information threshold when corrective examples are available; it does not make observability an identification guarantee.

At $r = h$, executing sixteen actions instead of one increases MSE by factors 7.82, 4.56, 2.63, and 1.01 as n grows. Longer execution delays correction. Noiseless fits tell a different story: coupled $n = 2$ is stable at history two but about 289 times less accurate than the noisy-data fit, while the original pair remains unstable. Thus the same ambiguity property does not force the same ridge-selected controller. Higher-dimensional comparisons use the coupled-family $n = 2$ baseline consistently.

The experiments do not verify Proposition 2's Lipschitz assumptions or its sufficient bound, which cannot certify $(r, \ell) = (16, 1)$. They also do not establish robustness to continuing test noise. Training amplitudes differ from test amplitudes, and dimension changes both information delay and the number of unexcited directions. Seed ranges are descriptive, not confidence intervals.

References

M. Simchowitz, D. Pfrommer, and A. Jadbabaie. The pitfalls of imitation learning when actions are continuous. *arXiv:2503.09722*, 2025.

M. Laskey, J. Lee, R. Fox, A. Dragan, and K. Goldberg. DART: Noise injection for robust imitation learning. *CoRL*, 2017.

S. Ross, G. Gordon, and J. A. Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. *AISTATS*, 2011.

T. T. Zhang, D. Pfrommer, C. Pan, N. Matni, and M. Simchowitz. Action chunking and exploratory data collection yield exponential improvements in behavior cloning for continuous control. *arXiv:2507.09061*, 2025.